HALO:

High-efficiency Autonomous Low-SWaP Operations

Sloan Hatter, Blake Gisclair

Advisor: Dr. Ryan T. White

MILESTONE 3 PROGRESS MATRIX

Task	Completion %	Sloan	To Do
Hardware Swap	100%	Switch out Pi for Jetson	None
Post Processing Scripts for Raspberry Pi	NULL	NULL	NULL
Post Processing Scripts for Jetson	90%	Modify	Modify available post processing scripts if needed.
8-bit Representation	50%	Implement	Implement the INT8 quantization methods.

TASKS COMPLETED

- Hardware Swap
  - Raspberry Pi 5 AI HAT+ → Jetson AGX Orin
- Post-Processing Scripts for Raspberry Pi
  - Pi scripts would not work on Jetson, so they have been discarded
- Post-Processing Scripts for Jetson
  - Post-Processing scripts are available, I would only need to modify them if they don’t provide proper analysis or to optimize performance
- 8-bit Representation
  - Utilize the NVIDIA TensorRT INT8 package from NVIDIA JetPack SDK

DEMO

MILESTONE 4 TASK MATRIX

Task	Sloan
4-bit Representation	100%
Tune 4-bit Model	100%
Research algorithms for binary quantization	100%

MILESTONE 4 TASKS

- 4-bit Representation
  - Utilize the NVIDIA TensorRT INT4 package from NVIDIA JetPack SDK
- Tune 4-bit Model
  - Utilize the Quantized-Aware Training method from the NVIDIA TensorRT package
- Research Algorithms for Binary Quantization
  - Research 1-bit quantization algorithms that focus on recouping accuracy loss

THANK YOU!
QUESTIONS?